Within-Generation and Transgenerational Plasticity of a Temperate Salmonid in Response to Thermal Acclimation and Acute Temperature Stress

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Accepted 6/15/2022; Electronically Published 9/26/2022

Online enhancement: appendix.

ABSTRACT

The rise in temperature associated with climate change may threaten the persistence of stenothermal organisms with limited capacities for beneficial thermal acclimation. We investigated the capacity for within-generation and transgenerational thermal responses in brook trout (*Salvelinus fontinalis*), a cold-adapted salmonid. Adult fish were acclimated to temperatures within (10°C) and above (21°C) their thermal optimum for 6 mo before spawning, then mated in a full factorial breeding design to produce offspring from cold- and warm-acclimated parents and bidirectional crosses between parents from both temperature treatments. Offspring from families were subdivided and reared at two acclimation temperatures representing their current (15°C) and anticipated future (19°C) habitat temperatures. Offspring thermal physiology was measured as the rate of oxygen consumption ($\dot{M}O_2$) during an acute change in temperature (increase of 2°C h^{-1}) to observe their $\dot{M}O_2$-temperature relationship. We recorded resting $\dot{M}O_2$, peak (highest achieved, thermally induced) $\dot{M}O_2$, and critical thermal maximum (CTM) as performance metrics. Although limited, within-generation plasticity was greater than transgenerational plasticity, with offspring warm acclimation elevating CTM by 0.5°C but slightly lowering peak thermally induced $\dot{M}O_2$. Transgenerational plasticity was evident as a slightly elevated resting $\dot{M}O_2$ and a shift of the $\dot{M}O_2$-temperature relationship to higher rates overall in offspring from warm-acclimated parents. Furthermore, offspring whose parents were warm acclimated were in worse condition than those whose parents were cold acclimated. Both parents contributed to offspring thermal responses; however, the paternal effect was stronger. Despite the existence of within-generation and transgenerational plasticity in brook trout, it is unlikely that these will be sufficient for coping with long-term changes to environmental temperatures.

Keywords: climate change, brook trout, metabolic rate, thermal tolerance, transgenerational acclimation, plasticity.

Introduction

Environmental warming as a result of climate change is adversely affecting the physiology and persistence of many species and populations globally (Moritz and Agudo 2013; Whitney et al. 2016). Species or populations that cannot migrate or are restricted to localized habitats are particularly vulnerable because they would be left to face temperatures higher than those they are physiologically capable of withstanding long-term (Somero 2010). Evolutionary change may provide the best option for long-term persistence for organisms; however, the accelerated rate of climate change is likely too rapid for most organisms to respond (Comte and Olden 2017). This is especially true for those with long generation times or limited standing genetic variation (Willi et al. 2006; Munday et al. 2013; Meier et al. 2014).

Thermal acclimation (phenotypic plasticity) may help to buffer temperature effects through physiological adjustments that can occur within a single generation or over multiple generations, potentially allowing some populations to compensate for short-term (within-generation) or long-term (transgenerational) environmental change (Jablonka et al. 1992; Somero 2010; Bonduriansky et al. 2012; Schulte 2015; Norouzitallab et al. 2019). Plasticity (within-generation and transgenerational) is thought to have evolved in populations that experience environmental variation over time (Leimar and McNamara 2015; Beaman et al. 2016), such as populations living in temperate regions. Here, we use the definition of transgenerational plasticity given in Bell and Hellmann (2019) and Bonduriansky (2021), which describe it as a form of plasticity where phenotypic changes occur over multiple

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  Published by The University of Chicago Press. https://doi.org/10.1086/721478